Linear Algebra.

Rank of a matrix by elimentary row transformations

The following are the elimentary row transformations (operations of a matrix). The transformations can also be applied for columns.

i) Interchange any two rows
ii) Multiplication any row by a non-zero const.
iii) Addition to any row a constant multiple of any other row.

iv) Elementy transformations along with the notation we use in the following table considring matrix.

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

Elmentary row transformation	Notations	Resultand matrix
i) Interchange first row & second row	$R_1 \leftrightarrow R_2$	$\begin{bmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$.
ii) Multiplication of third row by const k	$R_3 \to kR_3$	$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ kc_1 & kc_2 & kc_3 \end{bmatrix}$
iii) Addition to second row k times the first row.	$R_2 \to R_2 + kR_1$	$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1+ka_1 & b_1+ka_2 & b_3+ka_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$.

Echelon form of a matrix.

* A non-zero matrix A is said to be in row echelon form if it satifies the following conditions

i) All zero rows below the non-zero rows.

ii) The first non-zero entry in any non-zero row is 1.

eg $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 6 & 7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Rank of a matrix:

The rank A in its echelon form is equal to no. of non-zero rows. It is denoted by $\rho(A)$

Problems:

i) Find the Rank of the following matrixes by elementary row transformations

Ans. $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ $R_1 \to R_2 \to 2R_1$, $R_3 \to R_3 - R_1$

$\sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 2 \end{bmatrix} R_3 \to R_3 + R_2$

$\sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_2 \to -R_2$.

is in echelon form.

Thus the matrix A is having 2 non zero rows

Rank of matrix $A = \rho(A) = 2$.

Find the rank of the matrix.

$$A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} R_1 \leftrightarrow R_2.$$

$$A = \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} R_3 \longrightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix} R_3 \longrightarrow R_3 - R_2.$$

$$\sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 \longrightarrow \frac{1}{2} R_1 \\ R_2 \longrightarrow \frac{1}{2} R_2. \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 3/2 & 5/2 & 2 \\ 0 & 1 & 3/2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ is in echelon form.

∴ Thus matrix A is having 2-non zero rows Rank of matrix $A = \rho(A) = 2$.

$$A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Consider $$A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ -1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} R_1 \leftrightarrow R_2$$

$$A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & -1 & -3 & -1 \\ -1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} \begin{matrix} R_2 \longrightarrow R_2 - 2R_1 \\ R_3 \longrightarrow R_3 - R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & -5 & -9 & 1 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix} \quad R_2 \longleftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -2 & -2 & 2 \\ 0 & -5 & -9 & 1 \end{bmatrix} \quad \begin{matrix} R_3 \to R_3 + 2R_2 \\ R_4 \to R_4 + 5R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & -4 \end{bmatrix} \quad R_3 \longleftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 \to R_3/-4$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ It is in echelon form

$\therefore$ Thus the matrix A is having 3 non-zero rows so $A = \rho(A) = 3$.

$$A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$

Consider $A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$ $R_1 \longleftrightarrow R_2$.

$$A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 4 & 0 & 2 & 1 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$$ ~~$R_3 \longleftrightarrow R_1$~~

$R_2 \longrightarrow R_2 - 2R_1$,
$R_3 \longrightarrow R_3 - R_1$,
$R_4 \longrightarrow R_4 - R_1$,

~~$A = \begin{bmatrix} 2 & 3 & 4 & 7 \\ 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix}$~~

$$A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & -2 & 0 \end{bmatrix}$$ $R_3 \longrightarrow R_3 + R_2$
$R_4 \longrightarrow R_4 + R_2$.

$$A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & -6 & -7 \end{bmatrix}$$ $R_4 \longrightarrow R_4 - 2R_3$

$$A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & -2 & -4 & -7 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$ $R_1 \longrightarrow \frac{1}{2} R_1$
$R_2 \longrightarrow -\frac{1}{2} R_2$
$R_3 \longrightarrow -\frac{1}{3} R_3$

$$A = \begin{bmatrix} 1 & 1/2 & 3/2 & 2 \\ 0 & 1 & 2 & 7/2 \\ 0 & 0 & 1 & 4/3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$ is in echelon form.

$\rho(A) = 3$.

v) $$A = \begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad R_1 \longleftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} R_2 \longleftrightarrow R_2 + 2R_1 \\ R_3 \longleftrightarrow R_3 - R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 3 & 3 & 1 \\ 0 & -2 & -2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} R_2 \longleftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & -2 & -2 & 0 \\ 0 & 3 & 3 & 1 \end{bmatrix} \begin{matrix} R_3 \longleftrightarrow R_3 + R_2 \\ R_4 \longrightarrow R_4 - 3R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix} R_4 \longrightarrow R_4 + R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3 \longrightarrow R_3 \times 1/2$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ is in echelon form.

$\rho(A) = 3$.

H.W

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}.$$

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Consider $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} R_1 \longleftrightarrow R_2$

$$\approx \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} \begin{matrix} R_3 = R_3 - 3R_1, \\ R_4 = R_4 - R_1 \end{matrix}$$

$$A \approx \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus $f(A) = 2$.

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Consider $A \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} R_1 \longleftrightarrow R_2$

$$\approx \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \begin{matrix} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1, \\ R_4 = R_4 - (R_1 + R_2 + R_3) \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 = 5R_3 - 4R_2$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} R_2 = 1/5\, R_2 \\ R_3 = 1/33\, R_3 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 0 & 1 & 22/33 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus $\rho(A) = 3$.

Solution of system of linear equations.

Consider a system of 3 equations in 3 unknowns as

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}\right\} \quad \text{---- (1).}$$

Where a_1, a_2, a_3 are know coeffiencits

b_1, b_2, b_3 are know constants

x_1, x_2, x_3 are unknows to be determined.

$$\begin{aligned} 2x_1 + 7x_2 + 6x_3 &= 7. \\ 8x_1 + 6x_2 + 2x_3 &= 9. \\ x_1 + x_2 + x_3 &= 10. \end{aligned}$$

The above system of eqs can be written in the form of matrix as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

$$\Rightarrow Ax = B.$$

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is an column matrix of unknowns to ~~matrix~~ to be determined.

$B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ is a coloumn matrix of known constants.

Argmented matrix

We consider a new matrix comprising all the elements of matrix A. Along with the elements of the column matrix B. Such a matrix denoted by [A/B] is called the Argmented matrix.

(i.e) $[A/B] = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$.

The system of linear equations given in (1) can be solved by using following 2 methods.

i) Direct method (elimination method).

which gives the exact solutions to the system of linear equations after finite no. of steps. These methods are i) Gauss elimination method.

ii) Gauss - Jordon method.

iii) LU - decompostion method.

ii) Indirect method (iterative methods).

which gives the sequence of approx solutions & they converges to the exact solutions as the no. of steps terne to infinity.

Gauss elimination method.

Gauss elimination method is an direct method which gives the exact solutions to the given system of linear equations.

In this method we reduce the coefficient matrix A to equivalant upper triangular matrix U.

Then by gauss substitution we get exact solution values of given system of equations.

(i.e) $[A/B] \xrightarrow{GEM} [U/B']$.

Problems

1) Solve the following system of linear equations by gauss elimation method.

$$x + y + z = 9$$
$$x - 2y + 3z = 8.$$
$$2x + y - z = 3.$$

∴ We have augmented matrix.

$$[A/B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 1 & -2 & 3 & 8 \\ 2 & 1 & -1 & 3 \end{array}\right] \begin{array}{l} \\ R_2 \to R_2 - R_1, \\ R_3 \to R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -3 & -2 & -1 \\ 0 & -1 & -3 & -15 \end{array}\right] R_3 \to 3R_3 - R_2.$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -3 & -2 & -1 \\ 0 & 0 & -11 & -44 \end{array}\right] = [U/B'].$$

→ $x+y+z=9$

$-3y+2z=-1$

$-11z=-44$; $-11z=44$; $z=4$.

$-3y+2z=-1$

$-3y+2(4)=-1$

$-3y+8=-1$

$-3y=-9$; $y=3$.

$x+y+z=9$

$x+3+4=9$

$x+7=9$

$x=2$.

ii) $2x_1+x_2+4x_3=12$

$4x_1+11x_2-x_3=33$

$8x_1-3x_2+2x_3=20$.

Ans. We have argumented matrix.

$$\left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 4 & 11 & -1 & 33 \\ 8 & -3 & 2 & 20 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 4R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 9 & -9 & 9 \\ 0 & -7 & -14 & -28 \end{array}\right] \begin{array}{l} R_2 \to 1/9\, R_2 \\ R_3 \to 1/7\, R_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & -2 & -4 \end{array}\right] R_3 \to R_3 + R_2.$$

$$\sim \left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & -3 & -3 \end{array}\right] = [U/B'].$$

$$\begin{aligned} 2x_1 + x_2 + 4x_3 &= 12 \\ x_2 - x_3 &= 1 \\ -3x_3 &= -3. \end{aligned}$$

Consider $-3x_3 = -3$; $x_3 = 1$.

Consider $x_2 - x_3 = 1$; $-1 = 1 - x_2$; $x_2 = 2$.

Consider $2x_1 + x_2 + 4x_3 = 12$; $2x_1 = 6$; $x_1 = 3$.

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 2. \\ 3x_1 - x_2 + 4x_3 &= 4. \\ 2x_1 + x_2 - 2x_3 &= 5. \end{aligned}$$

Augmented matrix is

$$[B] \sim \left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 3 & -1 & 4 & 4 \\ 2 & 1 & -2 & 5 \end{array}\right] \begin{array}{l} R_3 = R_2 - 3R_1 \\ R_3 = R_3 - 2R. \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 0 & 5 & -5 & -2 \\ 0 & 5 & -8 & 1 \end{array}\right] R_3 = R_3 - R_2.$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 0 & 5 & -5 & -2 \\ 0 & 0 & -3 & 3 \end{array}\right].$$

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 2. \\ 5x_2 - 5x_3 &= -2. \\ -3x_3 &= 3. \end{aligned}$$

$x_3 = -1$ | $5x_2 - 5(-1) = -2$ | $x_1 - 2(-7/5) + 3(-1) = 2$.

$5x_2 = -2 - 5 = -7$ | $x_1 + 14/5 - 3 = 2$

$x_2 = -7/5$ | $x_1 = 5 - 14/5 = \frac{25-14}{5} = 11/5$

Thus $x_1 = 11/5$, $x_2 = -7/5$ & $x_3 = -1$.

iv) $2x + 5y + 7z = 52$

$2x + y - z = 0$.

$x + y + z = 9$.

Am. Augmented matrix is

$$[A/B] = \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 2 & 1 & -1 & 0 \\ 1 & 1 & 1 & 9 \end{array}\right] \quad R_1 \leftrightarrow R_3.$$

$$\approx \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 1 & -1 & 0 \\ 2 & 5 & 7 & 52 \end{array}\right] \quad \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 2R_1 \end{array}$$

$$\approx \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -1 & -3 & -18 \\ 0 & 3 & 5 & 34 \end{array}\right] \quad R_3 = R_3 + 3R_2.$$

$$\approx \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -1 & -3 & -18 \\ 0 & 0 & -4 & -20 \end{array}\right]$$

$\Rightarrow x + y + z = 9.$

$-y - 3z = -18$

$-4z = -20.$

$z = 5$ | $-y - 15 = -18$ | $x + 3 + 5 = 9$

$-y = 15 - 18$ | $x = 9 - 8$

$-y = -3$ | $x = 1.$

$x_1 + x_2 + x_3 + x_4 = 2$

$2x_1 - x_2 + 2x_3 - x_4 = -5$

$3x_1 + 2x_2 + 3x_3 + 4x_4 = 7$

$x_1 - 2x_2 - 3x_3 + 2x_4 = 5$

Augmented matrix

$$[A|B] = \begin{bmatrix} 1 & 1 & 1 & 1 & 2 \\ 2 & -1 & 2 & -1 & -5 \\ 3 & 2 & 3 & 4 & 7 \\ 1 & -2 & -3 & 2 & 5 \end{bmatrix} \begin{matrix} \\ R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \\ R_4 = R_4 - R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & -3 & 0 & -3 & -9 \\ 0 & -1 & 0 & 1 & 1 \\ 0 & -3 & -4 & 1 & 3 \end{bmatrix} R_2 \to -1/3 R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 3 \\ 0 & -1 & 0 & 1 & 1 \\ 0 & -3 & -4 & 1 & 3 \end{bmatrix} \begin{matrix} \\ \\ R_3 = R_3 + R_2 \\ R_4 = R_4 + 3R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & -4 & 4 & 12 \end{bmatrix} R_3 \longleftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & -4 & 4 & 12 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix} R_3 = 1/4 R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & -1 & 1 & 3 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix}$$

$x_1 + x_2 + x_3 + x_4 = 2$

$x_2 + 0x_3 + x_4 = 3$

$-x_3 + x_4 = 3$

$2x_4 = 4$

$x_4 = 2$ | $-x_3 + 2 = 3$, $x_3 = -1$ | $x_2 + 0 + 2 = 3$, $x_2 = 1$ | $x_1 + 1 - 1 + 2 = 2$, $x_1 = 0$.

$x_1 = 0, \quad x_2 = 1, \quad x_3 = -1 \text{ & } x_4 = 2$

Gauss Jordan method :

It is an direct method it gives the exact solutions to the given system of linear eqs. In this method, we reduce the coefficient matrix A of the given system of linear eqns to an equavalant diagonal form of the matrix T. Then directly we get the solutions values of the given system of equations.

Problems :

Solve the following system of equations by gauss Jordan method.

i) $x+y+z = 9$

$x-2y+3z = 8.$

$2x+y-z = 3$

Ans. Consider Argumented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 1 & -2 & 3 & 8 \\ 2 & 1 & -1 & 3 \end{array}\right] \begin{array}{l} R_2 \longrightarrow R_2 - R_1 \\ R_3 \longrightarrow R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -3 & 2 & -1 \\ 0 & -1 & -3 & -15 \end{array}\right] R_2 \longleftrightarrow R_3.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -1 & -3 & -15 \\ 0 & -3 & 2 & -1 \end{array}\right] R_2 \longrightarrow -R_2.$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -1 & 3 & 15 \\ 0 & -3 & 2 & -1 \end{array}\right] \begin{array}{l} R_1 \to R_1 - R_2 \\ R_3 \to R_3 + 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & -6 \\ 0 & 1 & 3 & 15 \\ 0 & 0 & 11 & 44 \end{array}\right] R_3 \to \tfrac{1}{11} R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & -6 \\ 0 & 1 & 3 & 15 \\ 0 & 0 & 0 & 4 \end{array}\right] R_1 \to R_1 + 2R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right] = [D/B']$$

$x = 2$
$y = 3$
$z = 4$

$$\begin{aligned} 2x + 5y + 7z &= 52 \\ 2x + y - z &= 0 \\ x + y + z &= 9 \end{aligned}$$

We have augmented matrix

$$[A/B] = \left[\begin{array}{ccc|c} 2 & 5 & 7 & 52 \\ 2 & 1 & -1 & 0 \\ 1 & 1 & 1 & 9 \end{array}\right] R_1 \leftrightarrow R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 1 & -1 & 0 \\ 2 & 5 & 7 & 52 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 2R_1 \end{array}$$

$$\approx \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -4 & -3 & -18 \\ 0 & 3 & 5 & 34 \end{array}\right] \begin{array}{l} R_1 \to R_1 + R_2 \\ \\ R_3 \to R_3 + 3R_2. \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & -9 \\ 0 & -1 & -3 & -18 \\ 0 & 0 & -4 & -20 \end{array}\right] R_3 \to -1/4\, R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & -9 \\ 0 & -1 & -3 & -18 \\ 0 & 0 & 1 & 5 \end{array}\right] \begin{array}{l} R_1 \to R_1 + 2R_3 \\ R_2 \to R_2 + 3R_3. \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & 5 \end{array}\right] = [D/B'].$$

$x = 1$

$y = 3$

$z = 5.$

Gauss - Seidel method :

* This is a numerical - ittirative method giving approx solution.

This method can't be applied to all the sytem of equations.

The method is applicable when numerically large co-efficients are along the leading

agonal of the co-efficient matrix. Such system is called a digonally deminate ystem.

metimes we may have to rearrange the iven system of equation to meet this requirement this condition is satisfied, the solution exists the ittination courses will converge.

The method is illustrated for a system of 3-independent eq in 3-unknowns.

Consider the system of eq. $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$, $a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_1$, $a_{31}x_1 + a_{32}x_2 + a_{33}x_3$ -

The system is solved if.

$|a_{11}| > |a_{12}| + |a_{13}|$.

$|a_{22}| > |a_{21}| + |a_{23}|$

$|a_{33}| > |a_{31}| + |a_{32}|$

Then the system diagonally dominant.

The given system of eq can be written in the form

$x_1 = 1/a_{11}\ [b_1 - a_{12}x_2 - a_{13}x_3]$

$x_2 = 1/a_{22}\ [b_2 - a_{21}x_1 - a_{23}x_3]$

$x_3 = 1/a_{33}\ [b_3 - a_{31}x_1 - a_{32}x_2]$.

Taking intial approximatios as

$x_1^{(0)} = 0 \quad x_2^{(0)} = 0 \quad , \quad x_3^{(0)} = 0.$

1st iteration

$$x_1^{(1)} = 1/a_{11} \left[b_1 - a_{12} \}.0 - a_{13} \cdot 0 \right]$$

$$x_2^{(1)} = 1/a_{22} \left[b_2 - a_{21} \cdot x_1^{(1)} - a_{23} \cdot 0 \right].$$

$$x_3^{(1)} = 1/a_{33} \left[b_3 - a_{31} \, x^{(1)} - a_{32} x_2^{(1)} \right]$$

This completes first iteration.

This process is continued till we get the solutions to the desired degree of accuracy.

Problems:

i) Solve the following system of equation by gauss - seidel method.

a) $10x + y + z = 12$

Ans.

$x + 10y + z = 12$

$x + y + 10z = 12$

The given system of equations are diagonally dominent. & the eqs are put in the form

$x = 1/10 \left[12 - y - z \right]$

$y = 1/10 \left[12 - x - z \right]$

$z = 1/10 \left[12 - x - y \right].$

Taking initially

$x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$.

I^st iteration:

$x^{(1)} = 1/10\,[12 - 0 - 0] = 1.2$

$y^{(1)} = 1/10\,[12 - 1.2 - 0] = 1.08$

$z^{(1)} = 1/10\,[12 - 1.2 - 1.08] = 0.972$.

IInd interation:

$(x)^{(1)} = 1/10\,[12 - 1.08 - 0.972] = 0.9948$

$y^{(2)} = 1/10\,[12 - 0.9948 - 0.972] = 1.00332$.

$z^{(2)} = 1/10\,[12 - 0.9948 - 1.00332] = 1.000188$

3rd iteration:

$x^{(3)} = 1/10\,[2 - 1.00332 - 1.000188] = 0.9996$.

$y^{(3)} = 1/10\,[12 - 0.99965 - 1.000188] = 1.00002$

$z^{(3)} = 1/10\,[12 - 0.99965 - 1.00002] = 1.00003$.

IVth iteration:

$x^{(4)} = 1/10\,[12 - 1.00002 - 1.00003] = 0.999995 \approx 1$

~~$y^{(4)} = 1/10\,[12 - 1 - 1.00003]$~~

$y^{(4)} = 1/10\,(12 - 1 - 1.00003) = 0.999997 \approx 1.$

$z^{(4)} = 1/10\,[12 - 1 - 1] = 1.$

$\therefore x = 1,\ y = 1,\ z = 1.$

2) $x + y + 54z = 110$

$27x + 6y - z = 85$

$6x + 15y + 2z = 72.$

The given system of equations is not diagonally dominent & hence we have to first rearrange the given system of equations as follows.

$27x + 6y - z = 85 \quad (|27| > |6| + |-1|)$

$6x + 15y + 2z = 72 \quad (|15| > |6| + |2|)$

$x + y + 54z = 110. \quad (|54| > |1| + |1|)$

$x = 1/27\,[85 - 6y + z]$

$y = 1/15\,[72 - 6x - 2z]$

$z = 1/54\,[110 - x - y]$

Taking initially,

let $x^{(0)} = 0,\ y^{(0)} = 0,\ z^{(0)} = 0$ be the intial approximations.

1st iteration:

$x^{(1)} = 1/27\,[85 - 6(0) + 0] = 3.14815$

$y^{(1)} = 1/15\,[72 - 6(3.14815) - 2(0)] = 3.54074$

$z^{(1)} = 1/54 \left[110 - 3.14815 - 3.54074\right] = 1.91317.$

II iteration:

$x^{(2)} = 1/27 \left[85 - 6(3.54074) + 1.91317\right] = 2.43218$

$y^{(2)} = 1/15 \left[72 - 6(3.14815) - 2(1.91317)\right] = 3.57204$

$z^{(2)} = 1/54 \left[110 - 2.43218 - 3.57204\right] = 1.92585.$

III iteration:

$x^{(3)} = 1/27 \left[85 - 6(3.57204) + 1.92585\right] = 2.42569$

$y^{(3)} = 1/15 \left[72 - 6(2.43218) - 2(1.92585)\right] = 3.57294$

$z^{(3)} = 1/54 \left[110 - 2.42569 - 3.57294\right] = 1.92595.$

IV iteration

$x^{(4)} = 1/27 \left[85 - 6(3.57294) + 1.92595\right] = 2.42549.$

$y^{(4)} = 1/15 \left[72 - 6(2.42549) - 2(1.92595)\right] = 3.57301$

$z^{(4)} = 1/54 \left[110 - 2.42549 - 3.57301\right] = 1.92595.$

$x = 2.42549, \quad y = 3.57301, \quad z = 1.92595.$

3) $20x + y - 2z = 17$

$2x + 20y - z = -18$

$2x - 3y + 20z = 25$

Ans. The equations are diagonally dominant & hence we first write them in the following form.

$x = 1/20\,[17 - y + 2z]$

$y = 1/20\,[-18 - 3x + z]$

$z = 1/20\,[25 - 2x + 3y]$

Let $x^{(0)} = 0$; $y^{(0)} = 0$ & $z^{(0)} = 0$ be initial approximations.

First iteration:

$x^{(1)} = 1/20\,[17 - 0 - 0] = 17/20 = 0.85$

$y^{(1)} = 1/20\,[-18 - 3(0.85)] = -1.0275$

$z^{(1)} = 1/20\,[25 - 2(0.85) + 3(-1.0275)] = 1.0109$

Second iteration:

$x^{(2)} = 1/20\,[17 - (-1.0275) + 2(1.0109)] = 1.0025$

$y^{(2)} = 1/20\,[-18 - 3(1.0025) + 1.0109] = -0.9998$

$z^{(2)} = 1/20\,[25 - 2(1.0025) + 3(-0.9998)] = 0.9998$

Third iteration:

$x^{(3)} = 1/20\ [17 - (-0.9998) + 2(0.9998)] = 0.99997$

$y^{(3)} = 1/20\ [-18 - 3(0.99997) + 0.9998] = -1.0000085$

$z^{(3)} = 1/20\ [25 - 2(0.999997) + 3(-1.0000055)] = 1.0000022$

$x = 1, \quad y = -1 \ \& \ z = 1$ is required

Thus solution.

Eigen values & eigen vectors of a square matrix

Given a square matrix A, if their exists a scalar λ (real ~~no~~ (or) complex) ~~such that~~ and a non-zero column matrix 'X'. Such that $AX = \lambda X$ then λ is called a eigen value of A & X is called an eigen vector of A corresponding to eigan value λ.

If I is the unit matrix of the same order as that of A we have ~~I =~~ .

$X = IX$ & hence $AX = \lambda X$. ~~A~~

~~where~~ $AX = \lambda IX$

$AX - \lambda IX = 0$.

$[A - \lambda I][X] = [0]$, where $[0]$ is a null matrix.

Let us consider a null matrix of order 3

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \text{ and also } \lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

It can be easily seen that $[A - \lambda I][x] = [0$

Represents a set of homogenous equations in three unknowns ~~i.e, $a_1 - \lambda$~~

$$\begin{bmatrix} a_1 \to & a_2 & a_3 \\ b_1 & b_2 - \lambda & b_3 \\ c_1 & c_2 & c_3 - \lambda \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

i.e)
$$(a_1 - \lambda)x + a_2 y + a_3 z = 0$$
$$b_1 x + (b_2 - \lambda)y + b_3 z = 0.$$
$$c_1 x + c_2 y + (c_3 - \lambda)z = 0.$$

A non-trivial solution for the system of equati exists if the determinent of the co-efficiet matrix is zero. i.e, $\begin{vmatrix} a_1 - \lambda & a_2 & a_3 \\ b_1 & b_2 - \lambda & b_3 \\ c_1 & ~~b_2 - \lambda~~ & ~~b_3~~ c_3 - \lambda \end{vmatrix} = 0$ (or) $|A - = 0$

on expanding we get a cubic equation in λ Which is called the characteristic equation of A. (i.e ~~determinet~~ $|A - \lambda I| = 0$ is called characteristic equation of A.] The roots of the equation (or) hygen values which are also calle the hygen roots (or) characteristic roots.

For each value of λ their will be an eigen vector $x \neq 0$. which is also called characteri

vector.

Rule of crossmultiplication.

$a_1x + a_2y + a_3z = 0$

$b_1x + b_2y + b_3z = 0.$

Then the proportionate values of x, y, z (i.e., $x : y : z$) is given by $\frac{x}{\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}}$

$= \frac{z}{\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}}$

Problems!

1) Find the all the eigen values & corresponding eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

Sol: $|A - \lambda I| = 0.$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0.$$

The characteristic equation of A is

$|A - \lambda I| = 0$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0.$$

$= (8-\lambda)[(7-\lambda)(3-\lambda)-16] + 6[-6(3-\lambda)+8] + 2[24-2(7-\lambda)]$

$= (8-\lambda)[21-3\lambda-7\lambda+\lambda^2-16] + 6[-18+6\lambda+8] + 2[24-14+2\lambda]$

$= 0.$

$= (8-\lambda)[\lambda^2-10\lambda+5] + 6[6\lambda-10] + 2[10+2\lambda] = 0.$

$= 8\lambda^2-80\lambda+40-\lambda^3+10\lambda^2-5\lambda+36\lambda-60+20+4\lambda=0$

$-\lambda^3+[8+10]\lambda^2+[-80-5+36+4]\lambda = 0.$

$-\lambda^3+18\lambda^2-45\lambda = 0.$

$\lambda^3-18\lambda^2+45\lambda = 0.$

$\lambda[\lambda^2-18\lambda+45] = 0.$

$\lambda = 0 \quad \& \quad \lambda^2-18\lambda+45 = 0.$

$\lambda^2-15\lambda-3\lambda+45 = 0$

$\lambda(\lambda-15)-3(\lambda-15) = 0.$

$(\lambda-3)(\lambda-15) = 0.$

$\lambda-3 = 0 \quad \& \quad \lambda-15 = 0.$

$\lambda = 3 \quad \& \quad \lambda = 15.$

$\therefore \lambda = 0, 3, 15$ are the eigen values of given matrix A.

Now consider $[A-\lambda I][x] = [0]$.

i.e, $(8-\lambda)x - 6y + 2z = 0.$

$-6x + (7-\lambda)y - 4z = 0$

$2x - 4y + (3-\lambda)z = 0.$

Case-1 : When $\lambda = 0$, we have.

$$5x - 6y + 2z = 0 \quad \text{---} (1)$$
$$-6x + 7y - 4z = 0 \quad \text{---} (2)$$
$$2x - 4y + 3z = 0 \quad \text{---} (3).$$

Apply the rule of cross multiplication for (1) & (2)

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}}$$

$$\frac{x}{24-14} = \frac{-y}{-32+12} = \frac{z}{56-36}.$$

$$\frac{x}{10} = -y/20 = z/20$$

$$\frac{x}{1} = y/2 = z/2 \quad \Big| \quad \therefore x_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \text{ is a eigen vector corresponding to eigen value } \lambda = 0.$$

$\Rightarrow x = 1, \; y = 2, \; z = 2.$

Case-2 : When $\lambda = 3$, we have.

$$5x - 6y + 2z = 0 \quad \text{---} (4)$$
$$-6x + 4y - 4z = 0 \quad \text{---} (5)$$
$$2x - 4y + 0z = 0 \quad \text{---} (6)$$

Apply the rule of cross multiplication from (4) & (5)

$$\frac{x}{\begin{matrix} -6 & 2 \\ 4 & -4 \end{matrix}} = \frac{-y}{\begin{matrix} 5 & 2 \\ -6 & -4 \end{matrix}} = \frac{z}{\begin{matrix} 5 & -6 \\ -6 & 4 \end{matrix}}$$

$$\frac{x}{+24-8} = \frac{-y}{-20+12} = \frac{z}{20-36}.$$

$$\frac{x}{+56} = \frac{-y}{-8} = \frac{z}{56-16}$$

$$\frac{x}{16} = \frac{y}{8} = \frac{z}{-16}.$$

$\div$ by 8.

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{-2} \quad ; \quad x=2,\ y=1,\ z=-1.$$

Case-3 : When $\lambda = 15$.

$$(8-15)x - 6y + 2z = 0$$
$$-6x + (7-15)y - 4z = 0.$$
$$2x - 4y + (3-15)z = 0.$$

$$-7x - 6y + 2z = 0 \quad — (7)$$
$$-6x - 8y - 4z = 0 \quad — (8)$$
$$2x - 4y - 12z = 0. \quad — (9).$$

Cross multiplication for eq (7) & (8).

$$\frac{x}{-7 \quad -6} = \frac{y}{-6 \quad 2} = \frac{z}{}$$
$$-6 \quad -8 \quad -8 \quad -4$$

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ -8 & -4 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -7 & +2 \\ -6 & -4 \end{vmatrix}} = \frac{}{\begin{vmatrix} & \\ & \end{vmatrix}}$$

$$\frac{x}{|24+16|} = \frac{-y}{28+12} = \frac{z}{56-36}.$$

$$\frac{x}{40} = \frac{-y}{40} = \frac{z}{20}$$

$\div$ by 20.

$x=2$, $y=-2$, $z=1$.

eigen vector corresponding to eigen value $\lambda =$

2) Find all the given eigen values & the corresponding eigen vector for the matrix

$$A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$$

Ans. The characteristic equation of A is,

$$|A - I\lambda| = 0$$

i.e., $$\begin{vmatrix} (7-\lambda) & -2 & 0 \\ -2 & (6-\lambda) & -2 \\ 0 & -2 & (5-\lambda) \end{vmatrix} = 0$$

$$(7-\lambda)\{(6-\lambda)(5-\lambda)-4\} + 2\{-2(5-\lambda)-0\} + 0 = 0$$

$$(7-\lambda)\{30-5\lambda-6\lambda+\lambda^2-4\} + 2\{-10+2\lambda\} = 0.$$

$$(7-\lambda)[\lambda^2-11\lambda+26\} - 20 + 4\lambda = 0.$$

$$7\lambda^2 - 77\lambda + 182 - \lambda^3 + 11\lambda^2 - 26\lambda - 20 + 4\lambda = 0.$$

$$-\lambda^3 + 18\lambda^2 - 99\lambda + 162 = 0.$$

$$\lambda^3 - 18\lambda^2 + 99\lambda - 162 = 0.$$

Puting $\lambda = 3$, we have

$$27 - 16z + 297 - 162 = 324 - 324 = 0$$

Thus $\lambda = 3$ is a root by inspection

Now by synthetic division as follows

3	1	-18	99	-162
	0	3	-45	162
	1	-15	54	0

$$\Rightarrow \lambda^2 - 15\lambda + 54 = 0.$$

$$\lambda^2 - 6\lambda - 9\lambda + 54 = 0$$

$$\lambda(\lambda-6) - 9(\lambda-6) = 0.$$

$$(\lambda-6)(\lambda-9) = 0.$$

$$\lambda = 6 \quad \& \quad \lambda = 9.$$

$\therefore \lambda = 3, 6, 9$ are the eigen values of A.

Now consider $[A - I\lambda][X] = [0]$.

$$(7-\lambda)x - 2y + 0z = 0$$
$$-2x + (6-\lambda)y - 2z = 0$$
$$0x - 2y + (5-\lambda)z = 0$$

Case-III : Let $\lambda = 3$, The system of equations becom

$$4x - 2y + 0z = 0 \quad \text{—} (1)$$
$$-2x + 3y - 2z = 0 \quad \text{—} (2)$$
$$0x - 2y + 2z = 0 \quad \text{—} (3)$$

Solve (1) & (2) by the rule of cross multiplicati

$$\frac{x}{\begin{vmatrix} -2 & 0 \\ 3 & -2 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 4 & 0 \\ -2 & -2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & -2 \\ -2 & 0 \end{vmatrix}}$$

$$\Rightarrow \frac{x}{4} = \frac{-y}{-2} = \frac{z}{-4} \Rightarrow \frac{x}{4} = \frac{y}{2} = \frac{z}{-4} \Rightarrow \frac{x}{2} = \frac{y}{2} = \frac{z}{-2}.$$

$x = 2$, $y = 1$ & $z = -2$.

Thus the eigen vector X_2 corresponding th eigen values $\lambda = 6$ is $X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$.

Case-III : Let $\lambda = 9$. The system of equations becom

$$-2x - 2y + 0z = 0 \quad \text{—} (6)$$
$$-2x - 3y - 2z = 0 \quad \text{—} (7)$$
$$0x - 2y - 4z = 0 \quad \text{—} (8)$$

Solve (6) & (7) by the rule of cross multipli

$$\frac{x}{\begin{vmatrix} -2 & 0 \\ -3 & -2 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -2 & 0 \\ -2 & -2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & -2 \\ -2 & -3 \end{vmatrix}}$$

(or) $\frac{x}{2} = \frac{-y}{2} = \frac{z}{1}$

$\Rightarrow x = 2$, $y = -2$ & $z = 1$

Thus the eigen vector x_3 corresponding the eigen value $\lambda = 9$ is $X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$.

3) Find all the eigen roots and the eigen vectors of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.

Ans. The characteristic equation of A is

i.e., $\begin{vmatrix} (6-\lambda) & -2 & 2 \\ -2 & (3-\lambda) & -1 \\ 2 & -1 & (3-\lambda) \end{vmatrix} = 0.$

$(6-\lambda)[(3-\lambda)(3-\lambda)-1] + 2[-2(3-\lambda)+2] + 2[2-2(3-\lambda)] = 0$

$(6-\lambda)[9-3\lambda-3\lambda+\lambda^2-1] + 2[-6+2\lambda+2] + 2[2-6+2\lambda] = 0$

$(6-\lambda)[8-6\lambda+\lambda^2] - 12 + 4\lambda - 4 + 4\lambda = 0.$

$48 - 8\lambda - 36\lambda + 6\lambda^2 + 6\lambda^2 - \lambda^3 - 12 + 4\lambda - 4 + 4\lambda = 0.$

$-\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0.$

$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$

Putting $\lambda = 2$, we have.

$8 - 48 + 72 - 32 = 80 - 80 = 0.$

Thus $\lambda = 2$ is a root

Now by synthetic division method.

$\lambda = 2$	1	-12	36	-32
	0	2	20	32
	1	-10	16	0

$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$

$\lambda^2 - 2\lambda - 8\lambda + 16 = 0$

$\lambda(\lambda-2) - 8(\lambda-2) = 0$

$(\lambda-2)(\lambda-8) = 0.$

$\lambda = 2, 8$.

$\therefore \lambda = 2, 2, 8$ are the eigen values of A.

Now consider $[A - I\lambda][x] = [0]$

$$\Rightarrow \quad (6-\lambda)x - 2y + 2z = 0$$
$$-2x + (3-\lambda)y - z = 0$$
$$2x - y + (3-\lambda)z = 0.$$

Case-1 : Let $\lambda = 2$ and the corresponding equation are

$$4x - 2y + 2z = 0$$
$$-2x + y - z = 0$$
$$2x - y + z = 0$$

The above set of equations are all same a we have only one independent equations $2x - y + z = 0$ & hence we choose two variable arbitrary.

Let $z = k_1$ & $y = k_2$

$$\therefore x = \frac{(k_2 - k_1)}{2}$$

Thus the eigen vector x_1 corresponding the eigen value $\lambda = 2$ is $x_1 = \begin{bmatrix} \frac{k_2 - k_1}{2} \\ k_2 \\ k_1 \end{bmatrix}$.

Case-2 : Let $\lambda = 8$ & corresponding eq are

$$-2x - 2y + 2z = 0 \quad - (1)$$
$$-2x - 5y - 1z = 0 \quad - (2)$$
$$2x - 1y - 5z = 0 \quad - (3).$$

Solve ① & ② by the rule of cross multiplication

$$\frac{x}{\begin{vmatrix} -1 & 2 \\ -5 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -2 & 2 \\ -2 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & -2 \\ -2 & -5 \end{vmatrix}} \Rightarrow \frac{x}{12} = \frac{-y}{6} = \frac{z}{-6} \Rightarrow \frac{x}{2} = \frac{y}{-1} = \frac{z}{1}$$

$\Rightarrow x = 2, y = -1, z = 1.$

Thus the eigen vector X_2 corresponding the eigen value $\lambda = 8$ is $X_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$.

~~Radius power method~~ :

Reyleighs's power method.

This is a ~~ittrital~~ ittrative method to determine the largest eighen value & corresponing eighen vector of a square matrix.

* Find the largest eighen value & the corresponding eigin vector of the matrix. $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ & taking initial eigin vector as $[100]^T$. Carry out 4 ittration.

Am. Given $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ $X^{(0)} = [100]^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$AX^{(0)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4+0+0 \\ 2+0+0 \\ -2+0+0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}.$$

$$= 4 \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix} = \lambda_1 x^{(1)}$$

$$Ax^{(1)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 4+0.5+0.5 \\ 2(1)+3(0.5)-1(0.5) \\ -2(1)+1(0.5)+5(-0.5) \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 4 \\ -4 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 0.8 \\ -0.8 \end{bmatrix} = \lambda_2 x^{(2)}.$$

$$Ax^{(2)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ +0.93 \\ 0.93 \end{bmatrix} = \begin{bmatrix} 5.86 \\ 5.86 \\ -5.72 \end{bmatrix} = 5.86 \begin{bmatrix} 1 \\ 0.98 \\ -0.98 \end{bmatrix}$$

$$= \lambda_4 x^{(2)}.$$

$$Ax^{(4)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.98 \\ -0.98 \end{bmatrix} = \begin{bmatrix} 5.96 \\ 5.92 \\ -5.92 \end{bmatrix} = 5.96 \begin{bmatrix} 1 \\ 0.99 \\ -0.99 \end{bmatrix} = \lambda$$

then conclude that the largest eigen value is 5.96. & corresponding eigen vector is

$$\begin{bmatrix} 1 \\ 0.99 \\ -0.99 \end{bmatrix}.$$

Find the dominent eighen value ad the corresponding eighen vector of matrix A

$= \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by rayleigh's power method, taking initial eigen vector as $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}^T$.

Given $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ $x^{(0)} = [1\ 1\ 1]^T = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$Ax^{(0)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \lambda_1 x^{(1)}.$$

$$Ax^{(1)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \lambda_2 x^{(2)}.$$

$$Ax^{(2)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix}$$

Note: Taking the largest element in modulus as a common factor.

$$Ax^{(2)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 0.75 \\ -1 \\ 0.75 \end{bmatrix} = \lambda_3 x^{(3)}.$$

$$Ax^{(3)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.75 \\ -1 \\ 0.75 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -3.5 \\ 2.5 \end{bmatrix} = 3.5 \begin{bmatrix} 0.71 \\ -1 \\ 0.71 \end{bmatrix}$$

$= \lambda_4 x^{(4)}$

$$A x^{(4)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.71 \\ -1 \\ 0.71 \end{bmatrix} = \begin{bmatrix} 2.42 \\ -3.42 \\ 2.42 \end{bmatrix} = 3.42 \begin{bmatrix} 0.7 \\ 0 \\ 0.7 \end{bmatrix}$$

$$= \lambda_5 x^{(5)}.$$

∴ The dominet eigen vadue is 3.42 & corresponding eigher vector is 0.71.

* Use reyleighs power method numeraciully largest eighen value & corresponding eigen vectors of the modrix $X = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$. Taking intial value as $[100]^T$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Am. $$A x^{(0)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 1 \\ 2 \end{bmatrix} = 25 \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix}$$

$$A x^{(1)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \begin{bmatrix} 25.2 \\ 1.12 \\ 1.68 \end{bmatrix} = 25.2 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix}$$

$$A x^{(2)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = 25.18 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \lambda_3 x^{(3)}$$

1) Find the dominant eigen value and the corresponding eigen vector of the matrix, $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by power method, taking the initial eigen vector as $[1\ 1\ 1]^T$.

Ans $Ax^{(0)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6-2+2 \\ -2+3-1 \\ 2-1+3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 4 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 0 \\ 0.67 \end{bmatrix}$

$= \lambda_1 x^{(1)}$.

$Ax^{(1)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.67 \end{bmatrix} = \begin{bmatrix} 6-0+2(0.67) \\ -2+0-1(0.67) \\ 2-0+3(0.67) \end{bmatrix} = \begin{bmatrix} 7.34 \\ -2.67 \\ 4.01 \end{bmatrix}$

$= 7.34 \begin{bmatrix} 1 \\ -0.36 \\ 0.55 \end{bmatrix} = \lambda_2 x^{(2)}$.

$Ax^{(2)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.36 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 6+(-2)(-0.36)+2(0.55) \\ -2+3(-0.36)-1(0.55) \\ 2-1(-0.36)+3(0.55) \end{bmatrix}$

$= \begin{bmatrix} 7.82 \\ -3.63 \\ 4.01 \end{bmatrix}$

$= 7.82 \begin{bmatrix} 1 \\ -0.46 \\ 0.51 \end{bmatrix} = \lambda_3 x^{(3)}$.

$Ax^{(3)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.46 \\ 0.51 \end{bmatrix} = \begin{bmatrix} 7.94 \\ -3.89 \\ 3.99 \end{bmatrix} = 7.94 \begin{bmatrix} 1 \\ -0.49 \\ 0.50 \end{bmatrix}$

$= \lambda_4 x^{(4)}$.

$Ax^{(4)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.49 \\ 0.50 \end{bmatrix} = \begin{bmatrix} 7.98 \\ -3.97 \\ 3.99 \end{bmatrix} = 7.98 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix}$

$= \lambda_5 x^{(5)}$.

$$A x^{(5)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ 4 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \lambda_6 x^{(5)}$$

$$A x^{(6)} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ 4 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix} = \lambda_7 x^{(7)}$$

$\therefore$ $\lambda_6 = \lambda_7 = 8$ & $x^{(6)} = x^{(7)} = \begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix}$.

Thus dominant eigen value is 8 & the corresponding eigen vector is $\begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix}$.

Similarity of matrices and Diagonalisation of a square matrix.

Two square matrices A & B of the same order are said to be similar if there exists a non-singular matrix P such that $B = P^{-1}AP$. Here B is said to be similar to A.

Diagonalisation of a square matrix.

If A is a square matrix of order n having n linearly independent eigen vectors then there exists an nth order square matrix P such that $P^{-1}AP$ is a diagonal matrix. P is called the modal matrix.

Working procedure (Diagonalisation of 2×2 matrix).

* We find eigen values λ_1, λ_2 & corresponding eigen vectors X_1, X_2 for second matrix A.
* We form the modal matrix $P = [X_1, X_2]$ and compute P^{-1}.

Note: To get adjoint of a second order matrix principal diagonal terms are inter changed and signs of the other terms are changed.

Eg. $A = \begin{bmatrix} 2 & 3 \\ 5 & -6 \end{bmatrix}$, adj $A = \begin{bmatrix} -6 & -3 \\ -5 & 2 \end{bmatrix}$.

Problems:

i) Reduce the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ to the diagonal form & hence find A^4.

Ans Consider the gi

The characteristic equation of A is.

$$|A - \lambda I| = 0.$$

$$\begin{vmatrix} -1-\lambda & 3 \\ -2 & 4-\lambda \end{vmatrix} = 0.$$

$$(-1-\lambda)(4-\lambda) + 6 = 0.$$

$$-4 + \lambda - 4\lambda + \lambda^2 + 6 = 0.$$

$\lambda^2 - 3\lambda + 2 = 0.$

$\lambda^2 - 2\lambda - \lambda + 2 = 0.$

$\lambda(\lambda - 2) - (\lambda - 2) = 0$

$(\lambda - 1)(\lambda - 2) = 0$

$\lambda - 1 = 0$ & $\lambda - 2 = 0.$

$\lambda = 1$ & $\lambda = 2.$

$\therefore \lambda = 1, 2$ are the ~~given~~ eigen values of A.

Now consider $[A - \lambda I][x] = [0].$

$(-1 - \lambda)x + 3y = 0.$

$-2x + (4 - \lambda)y = 0.$

Case - i) When $\lambda = 1$, the system of equation becomes

$-2x + 3y = 0.$

$-2x + 3y = 0.$

Consider $-2x + 3y = 0.$

$2x = 3y.$

$\frac{x}{3} = \frac{y}{2}$

$x = 3$ & $y = 2.$

$x_1 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is the eigen vector corresponding to $\lambda =$

Case ii) When $\lambda = 2$, the system of equation bec

$-3x + 3y = 0.$

$-2x + 2y = 0.$

Consider $-3x + 3y = 0.$

$3x = 3y$

$x=1, y=1,$ $\quad x_2 = \begin{bmatrix} x \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is the eigen vector corresponding to $\lambda = 2$

Consider $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$.

Model matrix $P = [x_1, x_2]$.

$$P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}.$$

$$\text{adj} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}, \quad |P| = 3 - 2 = 1.$$

$$P^{-1} = \frac{1}{|P|} \text{adj } P = \frac{1}{1} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

Now consider $D = P^{-1}AP$

$$= \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}.$$

$$= \begin{bmatrix} -1+2 & 3-4 \\ 2-6 & -6+12 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}.$$

$$= \begin{bmatrix} 1 & -1 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}.$$

$$= \begin{bmatrix} 3-2 & 1-1 \\ -12+12 & -4 \end{bmatrix}.$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \text{Diag}(1, 2).$$

To find A^4

$$A^n = PD^nP^{-1}$$

$$A^4 = PD^4P^{-1} \text{ where } D^4 = \begin{bmatrix} 1^4 & 0 \\ 0 & 2^4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 16 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 16 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

$$\begin{bmatrix} 3+0 & 0+16 \\ 2+0 & 0+16 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 16 \\ 2 & 16 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3-22 & -3+48 \\ 2-32 & -2+4 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} -29 & 45 \\ -30 & 46 \end{bmatrix}$$

ii) Diagonalize matrix $\begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$

Am. Let $A = \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$

The characteristic equation of A is

$$|A - \lambda I| = 0.$$

$$\begin{vmatrix} -19-\lambda & 7 \\ -42 & 16-\lambda \end{vmatrix} = 0.$$

$(-19-\lambda)(16-\lambda)+(42)(7)=0$.

$-304-16\lambda+19\lambda+\lambda^2+294=0$.

$\lambda^2+3\lambda-10=0$.

$\lambda^2+5\lambda-2\lambda-10=0$.

$\lambda(\lambda+5)-2(\lambda+5)=0$

$(\lambda-2)(\lambda+5)=0$.

$\lambda-2=0$ & $\lambda+5=0 \Rightarrow \lambda=2$ & $\lambda=-5$.

$\therefore \lambda=2,-5$ are the eigen values of A.

Now consider $[A-\lambda I][x]=[0]$.

$$\begin{bmatrix} -19-\lambda & 7 \\ -42 & 16-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow (-19-\lambda)(x)+7y=0$

$-42x+(16-\lambda)y=0$.

Case-I : Let $\lambda=2$, we have.

$-21x+7y=0$

$-42x+14y=0$ (or) $-21x+7y=0$.

i.e, $-21x=-7y$.

$3x=y$

$x=y/3$

(or) $\frac{x}{1}=\frac{y}{3}$

~~$3x=y$~~

$x=1, y=3$.

$x_1=\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is the eigen vector correspon-ding to $\lambda=2$.

Case-2 : $\lambda=-5$, we have.

$-14x+7y=0$.

$-42x+21y=0$ (or) $-14x+7y=0$.

$14x = 7y$

$\frac{x}{1} = \frac{y}{2}$.

$x = 1$ & $y = 2$.

$x_2 = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is the eigen vector corresponding to eigen value $\lambda = -5$.

modal matrix $P = [x_1, x_2]$.

$$P = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}.$$

$$\text{adj } P = \begin{bmatrix} 2 & -1 \\ -3 & 1 \end{bmatrix}, |P| = 2 - 3 = 1.$$

$$P^{-1} = \frac{1}{|P|} \cdot \text{adj } P = \frac{1}{-1} \begin{bmatrix} 2 & -1 \\ -3 & 1 \end{bmatrix} = -\begin{bmatrix} 2 & -1 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}$$

Now $D = P^{-1}AP$.

$$= \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}.$$

$$= \begin{bmatrix} 38 - 42 & -14 + 16 \\ -57 + 42 & 21 - 16 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 2 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}.$$

$$= \begin{bmatrix} -4 + 6 & -4 + 4 \\ -15 + 15 & -15 + 10 \end{bmatrix}.$$

$D = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$ is the diagonal matrix.

Thus $P^{-1}AP = \text{Diag}(2, -5)$.